By making use of the uniform norm, we can obtain a necessary and sufficient condition for uniform convergence that is often useful.

8.1.10 Cauchy Criterion for Uniform Convergence *Let (f_n) be a sequence of bounded functions on $A \subseteq \mathbb{R}$. Then this sequence converges uniformly on A to a bounded function f if and only if for each $\varepsilon > 0$ there is a number $H(\varepsilon)$ in $\mathbb{N}$ such that for all $m, n \geq H(\varepsilon)$, then $||f_m - f_n||_A \leq \varepsilon$.* □

Proof. ($\Rightarrow$) If $f_n \rightrightarrows f$ on A, then given $\varepsilon > 0$ there exists a natural number $K(\frac{1}{2}\varepsilon)$ such that if $n \geq K(\frac{1}{2}\varepsilon)$ then $||f_n - f||_A \leq \frac{1}{2}\varepsilon$. Hence, if both $m, n \geq K(\frac{1}{2}\varepsilon)$, then we conclude that

$$|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f_n(x) - f(x)| \leq \tfrac{1}{2}\varepsilon + \tfrac{1}{2}\varepsilon = \varepsilon$$

for all $x \in A$. Therefore $||f_m - f_n||_A \leq \varepsilon$ for $m, n \geq K(\frac{1}{2}\varepsilon) =: H(\varepsilon)$.

($\Leftarrow$) Conversely, suppose that for $\varepsilon > 0$ there is $H(\varepsilon)$ such that if $m, n \geq H(\varepsilon)$, then $||f_m - f_n||_A \leq \varepsilon$. Therefore, for each $x \in A$ we have

$$|f_m(x) - f_n(x)| \leq ||f_m - f_n||_A \leq \varepsilon \qquad \text{for} \quad m, n \geq H(\varepsilon). \tag{8}$$

It follows that $(f_n(x))$ is a Cauchy sequence in $\mathbb{R}$; therefore, by Theorem 3.5.5, it is a convergent sequence. We define $f : A \to \mathbb{R}$ by

$$f(x) := \lim(f_n(x)) \qquad \text{for} \quad x \in A.$$

If we let $n \to \infty$ in (8), it follows from Theorem 3.2.6 that for each $x \in A$ we have

$$|f_m(x) - f(x)| \leq \varepsilon \qquad \text{for} \quad m \geq H(\varepsilon).$$

Therefore the sequence (f_n) converges uniformly on A to f. Q.E.D.

Exercises for Section 8.1

1. Show that $\lim(x/(x + n)) = 0$ for all $x \in \mathbb{R}, x \geq 0$.
2. Show that $\lim(nx/(1 + n^2x^2)) = 0$ for all $x \in \mathbb{R}$.
3. Evaluate $\lim(nx/(1 + nx))$ for $x \in \mathbb{R}$, $x \geq 0$.
4. Evaluate $\lim(x^n/(1 + x^n))$ for $x \in \mathbb{R}$, $x \geq 0$.
5. Evaluate $\lim((\sin nx)/(1 + nx))$ for $x \in \mathbb{R}$, $x \geq 0$.
6. Show that $\lim(\text{Arctan } nx) = (\pi/2)\text{sgn } x$ for $x \in \mathbb{R}$.
7. Evaluate $\lim(e^{-nx})$ for $x \in \mathbb{R}, x \geq 0$.
8. Show that $\lim(xe^{-nx}) = 0$ for $x \in \mathbb{R}$, $x \geq 0$.
9. Show that $\lim(x^2e^{-nx}) = 0$ and that $\lim(n^2x^2e^{-nx}) = 0$ for $x \in \mathbb{R}$, $x \geq 0$.
10. Show that $\lim((\cos \pi x)^{2n})$ exists for all $x \in \mathbb{R}$. What is its limit?
11. Show that if $a > 0$, then the convergence of the sequence in Exercise 1 is uniform on the interval $[0, a]$, but is not uniform on the interval $[0, \infty)$.
12. Show that if $a > 0$, then the convergence of the sequence in Exercise 2 is uniform on the interval $[a, \infty)$, but is not uniform on the interval $[0, \infty)$.
13. Show that if $a > 0$, then the convergence of the sequence in Exercise 3 is uniform on the interval $[a, \infty)$, but is not uniform on the interval $[0, \infty)$.

14. Show that if $0 < b < 1$, then the convergence of the sequence in Exercise 4 is uniform on the interval $[0, b]$, but is not uniform on the interval $[0, 1]$.
15. Show that if $a > 0$, then the convergence of the sequence in Exercise 5 is uniform on the interval $[a, \infty)$, but is not uniform on the interval $[0, \infty)$.
16. Show that if $a > 0$, then the convergence of the sequence in Exercise 6 is uniform on the interval $[a, \infty)$, but is not uniform on the interval $(0, \infty)$.
17. Show that if $a > 0$, then the convergence of the sequence in Exercise 7 is uniform on the interval $[a, \infty)$, but is not uniform on the interval $[0, \infty)$.
18. Show that the convergence of the sequence in Exercise 8 is uniform on $[0, \infty)$.
19. Show that the sequence (x^2e^{-nx}) converges uniformly on $[0, \infty)$.
20. Show that if $a > 0$, then the sequence $(n^2x^2e^{-nx})$ converges uniformly on the interval $[a, \infty)$, but that it does not converge uniformly on the interval $[0, \infty)$.
21. Show that if (f_n), (g_n) converge uniformly on the set A to f, g, respectively, then $(f_n + g_n)$ converges uniformly on A to $f + g$.
22. Show that if $f_n(x) := x + 1/n$ and $f(x) := x$ for $x \in \mathbb{R}$, then (f_n) converges uniformly on $\mathbb{R}$ to f, but the sequence (f_n^2) does not converge uniformly on $\mathbb{R}$. (Thus the product of uniformly convergent sequences of functions may not converge uniformly.)
23. Let (f_n), (g_n) be sequences of bounded functions on A that converge uniformly on A to f, g, respectively. Show that (f_ng_n) converges uniformly on A to fg.
24. Let (f_n) be a sequence of functions that converges uniformly to f on A and that satisfies $|f_n(x)| \leq M$ for all $n \in \mathbb{N}$ and all $x \in A$. If g is continuous on the interval $[-M, M]$, show that the sequence $(g \circ f_n)$ converges uniformly to $g \circ f$ on A.

Section 8.2 Interchange of Limits

It is often useful to know whether the limit of a sequence of functions is a continuous function, a differentiable function, or a Riemann integrable function. Unfortunately, it is not always the case that the limit of a sequence of functions possesses these useful properties.

8.2.1 Examples **(a)** Let $g_n(x) := x^n$ for $x \in [0, 1]$ and $n \in \mathbb{N}$. Then, as we have noted in Example 8.1.2(b), the sequence (g_n) converges pointwise to the function

$$g(x) := \begin{cases} 0 & \text{for } 0 \leq x < 1, \\ 1 & \text{for } x = 1. \end{cases}$$

Although all of the functions g_n are continuous at $x = 1$, the limit function g is not continuous at $x = 1$. Recall that it was shown in Example 8.1.6(b) that this sequence does not converge uniformly to g on $[0, 1]$.

(b) Each of the functions $g_n(x) = x^n$ in part (a) has a continuous derivative on $[0,1]$. However, the limit function g does not have a derivative at $x = 1$, since it is not continuous at that point.

(c) Let $f_n : [0, 1] \to \mathbb{R}$ be defined for $n \geq 2$ by

$$f_n(x) := \begin{cases} n^2x & \text{for } 0 \leq x \leq 1/n, \\ -n^2(x - 2/n) & \text{for } 1/n \leq x \leq 2/n, \\ 0 & \text{for } 2/n \leq x \leq 1. \end{cases}$$